

1 Dijkstra's

- 1.1 In an ocean, there could be multiple shortest paths between two whales. Given a Group of Whales, G , that are all equally spaced apart from each other in integer amounts and can travel these currents in either direction (which can be thought of as edges) with a starting Whale of Stella s and a target whale Tim t , devise an algorithm that returns the shortest path from s to t that uses the fewest number of edges. You can assume that there will be no ties.

Hint: Consider how you can modify the graph and run a shortest paths algorithm.

2 Maximal Spanning Trees

2.1 We have a graph that represents the social network of everyone in UPE, the LS CS Honor Society. The undirected edge weights correspond to an integer between 1 and 100 that states how well the 2 people (denoted by vertices) know each other! Consider the problem of finding a Maximum Spanning Tree.

(a) Describe a modification to our graph such that Kruskal's algorithm would allow us to find a Maximum Spanning Tree of a graph.

(b) Can we use the same modification approach to our graph such that Dijkstra's algorithm finds the Maximum Path between two UPE members?

3 UPE

- 3.1 You are the President of Berkeley's UPE Chapter! In order to manage your candidates, you have appointed officers to schedule 1-on-1s with candidates. Every officer can govern their candidates as well as any candidates that are friends of their candidate. Your job as President is to figure out the optimal way to allocate officers and build connections between candidates.

Formally, consider a graph G with vertices V and edges E . Each vertex v represents a candidate. It has an associated cost c , the cost of installing an officer in the town. Each edge e represents a potential connection. It has an edge weight w , the cost of building that connection (in units of time). Devise an algorithm that can efficiently compute which candidates to schedule 1-on-1s with which officers as well as which connections to form, such that every candidate in UPE has an officer that manages them (either has an officer to 1-on-1 with or is connected by friends to someone with a 1-on-1 to an officer).

4 Trie-ing to be Whale-fully efficient

- 4.1 Given a list of Professional Events (possibly repeated), devise a strategy to efficiently return a list of all the words that start with a given prefix.

5 Whalespaper

- 5.1 Given a dictionary of n words, describe a procedure for checking if a new compound word of length l can be created out of the concatenation of two words in the dictionary. For example, if our dictionary contains the words, "news", "paper", "new", and "whales", given a new word: "newspaper" check whether or not it is the concatenation of any two words in our dictionary. Give the runtime for your answer and explain why.